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1. Title of the invention:

A System and Method for a Destination-Assigned Allocation-Lifecycle Grant Identity Independent of Host-Visible Address and Independent of Address Translation, with Source-Fail-Closed Per-Nonce Terminal Write Tickets and Destination Drain Before Address Reuse

2. Field /Area of invention:

The invention pertains to the fields of:

- Computer architecture and memory-system security
- Fabric-attached, device-attached, pooled, and disaggregated memory systems, including CXL- and PCIe-connected memory
- Destination-side allocation-lifecycle identity maintained separately from the host-visible address, the device physical address, a link-security key, and the address-translation result
- Source-side fail-closed control of whether a memory operation is placed on the fabric or marked committable
- Destination-side drain before the same host-visible address is bound to a successor grant identity

The invention is not a method of doing business, a cryptographic handshake, a two-phase commit protocol, an address-translation service, a link-encryption scheme, or a rule that stale DMA must be prevented. Fabric-attached memory is the field of use. The invention is a destination-assigned allocation-lifecycle grant identity and a source-fail-closed write-authorization gate in a memory system.

3. Prior Patents and Publications from literature (provide a table summarizing the prior art)

Literature survey, 18 September 2026. Official titles and assignees are taken from the cited product pages, specifications, or Google Patents records. Not an InPASS / USPTO attorney search.

3.1 Prior art summary table

#	Reference	Year	What it does	Key limitation (gap this invention addresses)
1	PCI-SIG Address Translation Services (ATS) — Translation Request / Translation Completion (TReq/TCpl), PCIe Base Specification	—	A translation agent returns a cached, multi-use TCpl. Later memory requests may use that translation without a new completion.	Cached multi-use translation is not an authorized write. A remaining-count on one ticket is ATS. Claimed tickets are consume-once; each nonce N is terminal after commit, abort, or expire.

2	US 10,949,358 B2 — <i>Secure address translation services using message authentication codes and invalidation tracking</i> (Intel Corp., granted 16 March 2021); related US 11,921,646 B2 — <i>Secure address translation services using a permission table</i> ; US 11,899,593 B2 — <i>Method and apparatus for detecting ATS-based DMA attack</i>	2021 / 2024	IOMMU authenticates a device-supplied host physical address with a MAC and an invalidation-tracking table (ITT), or a host permission table; related art detects ATS-based DMA misuse.	Control is keyed by HPA (or a permission table on the address). After reuse of the same HPA for a new allocation, an old MAC/HPA can still look valid. Grant identity G is not the address and not an epoch on the address.
3	Gen-Z Consortium — Gen-Z Core Specification, Region Key (R-Key)	2020	A Fabric Manager installs a reusable opaque page key on requester and responder. The responder checks the key. The specification states the R-Key is not a security mechanism.	FM-installed reusable page key. Destination does not assign allocation-lifecycle G, does not issue consume-once tickets, and does not drain before a successor grant at the same host-visible address.
4	Space-Control — <i>Process-Level Isolation for Sharing CXL-based Disaggregated Memory</i> (arXiv:2603.06951)	2026	Fabric Manager issues a reusable public authentication label L_exp for a process context. A permission checker validates tagged loads/stores against a permission table.	FM-issued reusable range/process label. Not dest-issued, not consume-once, not a dest reuse barrier at the same host-visible address independent of that address.
5	Mao et al. — <i>WriteGuards: Distributed Storage Support for Strongly Consistent Caches</i> (OSDI 2026)	2026	Each write carries a fencing value tied to the current key-range owner. Storage rejects delayed writes after ownership change. Source still issues the write.	Destination/storage only checks a fence. It does not issue a consume-once ticket. Source is not fail-closed on placing the memory operation. If G is only an epoch on the address, this distinction fails.
6	CXL Dynamic Capacity Device (DCD) tag UUID; Linux 2026 DCD RFC (tag-aware uuid claim / seq_num); CXL Trusted Security Protocol (TSP) TE-state reset on add	2024 / 2026	Tag UUID groups mailbox extents; seq_num orders a tagged group. TE-state reset on add clears residue on the range. Address is not treated as grant identity.	Tag UUID is grouping metadata, not a per-commit ticket and not dest-local G. TE-state reset does not say the range is the same grant. Extent-list generation is an epoch on capacity, not dest-assigned allocation-lifecycle identity.

7	CVE-2025-9614 Delayed Posted Redirection (PCI-SIG PCIe IDE / TDISP specification gap; Intel advisory)	2025	Posted writes delayed across TDI rebinding may be processed in a new trusted domain. Official mitigation is flush outstanding posted transactions and refresh (re-key) the IDE stream before rebind.	IDE re-key / flush protects the link/session, not dest- assigned grant identity. Re-key is not dest-local G, not consume-once write tickets, and not Outstanding(G)=0 before address reuse.
8	CN122513106A — <i>A method and system for safe management and rapid revocation of CXL device P2P access</i> (Huazhong University of Science and Technology; pub. 4 August 2026; pending)	2026	CXL P2P security token carrying dest id, authorized address range, nonce, epoch, and revocation version VN. Destination may reject a reused nonce or an old VN.	Token is bound to a range , not dest- assigned G independent of address. Source fail- closed on the memory transaction is not this reference. Dest will- not-bind a successor G at the same host- visible address while Outstanding(G) is not zero is not this reference.
9	InfiniBand Trade Association — InfiniBand Architecture, Type-2 Memory Window and SEND_WITH_INV	—	Destination invalidates an R_Key after a send; the window may be rebound with a new key. The key is reusable until invalidate. Source still posts RDMA with the key.	Source gate is on a send-invalidate control path, not on the memory transaction. G is not an FM- or QP- installed reusable page key.
10	PCI-SIG Process Address Space ID (PASID) Stop Marker	—	A Stop Marker lets software reallocate a PASID after draining page requests of the old incarnation.	Same <i>shape</i> as G → DRAINING → G- prime when Outstanding(G)=0, but it is page-request identity reuse, not dest-issued write tickets on fabric stores. If G is only an incarnation of the address, this distinction fails.
11	US 10,185,680 B2 — <i>Secure direct memory access</i> (Intel Corp., granted 22 January 2019)	2019	Destination range registers are established for one DMA operation, then cleared. Writes are allowed only inside that start/end range.	Per-operation dest address range , not dest-local grant identity independent of address, and not source fail-closed fabric forward.

12	PCI-SIG TEE Device Interface Security Protocol (TDISP) / TEE-IO	2022	DMA is allowed while the TDI is RUN on a bound IDE stream. Configuration lock binds the interface to a trusted domain.	Session bind, not consume-once dest-issued tickets. TDISP configuration lock is not dest-assigned allocation-lifecycle G.
13	CXL 3.0 Global Fabric Attached Memory (GFAM) translation-table ranges; Stark, Marketos, and Moore — <i>How Flexible Is CXL's Memory Protection?</i> (ACM Queue / Communications of the ACM, 2023; DOI 10.1145/3606014); CapFabric signed capability-token model	2022 / 2023	GFAM programs up to eight FM translation-table ranges per endpoint. CACM discusses capability tokens with epoch revocation. CapFabric is a software model of signed HMAC tokens with epoch rotation.	FM-programmed reusable ranges or signed tokens with epoch revocation. Not dest-assigned G plus source no-forward of the memory op plus dest reuse barrier at the same HPA.
14	Alam et al. — <i>CryptoMMU: Enabling Scalable and Secure Access Control of Third-Party Accelerators</i> (MICRO 2023); Wang et al. — <i>DMAAUTH: A Lightweight Pointer Integrity-based Secure Architecture to Defeat DMA Attacks</i> (USENIX Security 2024); Ackermann et al. — <i>Northcape: Embedded Real-Time Capability-Based Addressing</i> (EuroS&PW 2024)	2023 / 2024	Host MAC over cached translations (CryptoMMU); bus authenticator with per-mapping identifier (DMAAUTH); northbridge capability check (Northcape).	Host MAC / bus PAC / mapping-id. DmaAuth's per-mapping identifier is an epoch-like mapping id, not dest-local G plus consume-once dest-issued tickets.
15	US 11,386,018 B2 — <i>Memory system and operating method thereof</i> (SK hynix Inc., granted 12 July 2022); JEDEC NVDIMM / ACPI NFIT write-persistence credits; CXL Integrity and Data Encryption (IDE)	2022 / —	NAND target-area nonce derived from physical address; NVDIMM write-buffer retire/credits; link encryption and integrity on the interconnect.	Area nonce, write-buffer credit, or link crypto. None is dest-assigned allocation-lifecycle grant identity, source-fail-closed per-nonce write tickets, and dest drain before address reuse.

3.2 Gap analysis

Existing approach	This invention
Cached ATS completion, or a remaining-count on one ticket	Consume-once per-nonce terminal write tickets; translation is not an authorized write
MAC / ITT / permission table keyed by host physical address	G is dest-assigned and is not the address; old tickets do not honor a new grant at the same HPA
Fabric-Manager reusable R-Key or L_exp	Destination assigns G; destination issues DCC; destination drains before a successor G at the same host-visible address
Destination/storage fence or fencing-value check; source still forwards (WriteGuards)	Destination issues tickets; source fail-closed does not place the memory operation without an unconsumed DCC
DCD tag UUID, TE-state reset, or IDE re-key / flush	Tag is grouping metadata; TE-state reset and IDE re-key are not grant identity
Nonce token bound to an address range (CN122513106A)	G is independent of address; source gate is on the memory transaction; dest will not bind G-prime while Outstanding(G) is not zero
SEND_WITH_INV / reusable Type-2 memory window	Gate is on the memory transaction, not send-invalidate; G is not a page key
PASID Stop Marker (page-request incarnation reuse)	Page-request identity, not dest-issued write tickets on fabric stores
Per-DMA destination address-range registers, then cleared	Dest-local grant identity independent of address, plus source fail-closed fabric forward

The present system is not ATS plus a remaining-count, and not an opaque key the destination only checks. Destination-assigned allocation-lifecycle grant identity G, source-fail-closed per-nonce terminal write tickets, and destination drain before address reuse are not disclosed as a combination in the surveyed art.

4.Summary and background of the invention (Address the gap / Novelty)

Patentable Subject Matter Statement

The present invention is **not** a mathematical method, **not** a business method, and **not** a computer programme per se within the meaning of **Section 3(k)** of the Patents Act, 1970 (read with the CRI Guidelines). It is **not** a method of doing business and is **not** claimed as “stale DMA must be prevented,” as a two-phase commit protocol, as an address-translation service, or as a link-encryption scheme. It is a **technical system** comprising a source boundary and a destination memory boundary, each comprising processing circuitry coupled to a memory and operatively coupled over a fabric or device interconnect, that performs **write-authorization isolation in a memory system**. The source boundary does or does not place a memory transaction on the fabric, or does or does not mark it committable. The destination boundary does or does not accept the operation as an **authorized write**, and does or does not bind a successor grant identity to the same host-visible address. Those observable control points — a class of memory transactions never placed on the fabric, never accepted as authorized writes, and never succeeded by a new grant at the same address while prior tickets remain outstanding — are the technical contribution.

Grant identity G, per-nonce ticket N, the tuple (C, Ain, G, N), and Outstanding(G) are **not** the product of the invention. They are **embedded control parameters of the gate** (whether the source may forward, whether the destination may accept an authorized write, and whether the destination may bind a successor grant). They serve a larger technical process: controlling write authorization when the same host-visible address is later bound to a new allocation. That is an improvement in the functioning of memory-system machinery, not a business rule.

Fabric-attached, device-attached, pooled, and disaggregated memory, including CXL- and PCIe-connected

memory, is only a **field of use**. The invention is a destination-assigned allocation-lifecycle grant identity and a source-fail-closed write-authorization gate.

Background

A memory system that reuses a host-visible address after an allocation is released must decide **what** is authorized at that address. Address Translation Services return a cached, multi-use translation completion. Link integrity and data encryption authenticate a stream, not an allocation. A Fabric Manager may install a reusable opaque page or range key. A destination may check a fencing value while the source still issues the write. A dynamic-capacity tag UUID groups mailbox extents; a TE-state reset clears residue on add; an IDE re-key flushes a stream on rebind. None of those mechanisms names the **allocation lifecycle** at the destination independently of the address bits and independently of the translation result.

After release and re-add at the same host-visible address, an old MAC over that address, an old cached translation, an old reusable key, or an old range-bound token can still look valid to a checker that keys authorization by address. The source can still place the corresponding memory transaction on the fabric. The destination can still accept a write whose address bits match the new backing. The problem the invention owns is that **the address is not the grant**. The invention is not a rule that delayed writes must be prevented.

A remaining-count on one reusable ticket is still a cached multi-use completion. A nonce bound to an address range is still keyed by the range. A Stop Marker that drains page-request incarnations of a PASID is page-request identity reuse, not destination-issued write tickets on fabric stores. These are properties of **memory-system authorization machinery**, not of a cryptographic handshake and not of a two-phase commit protocol.

The gap

1. Cached ATS completions, and a remaining-count on one ticket, authorize later memory requests without a new consume-once ticket; translation is not an authorized write.
2. MAC, invalidation-tracking, and permission-table checks keyed by host physical address still honor an old MAC or permission after the same HPA is bound to a new allocation.
3. Fabric-Manager reusable R-Key and L_exp labels are installed keys the destination only checks; they are not dest-assigned allocation-lifecycle identity and not consume-once dest-issued tickets.
4. A destination/storage fencing-value check (WriteGuards) rejects a delayed write after ownership change, but the source still issues the write and the destination does not issue a consume-once ticket.
5. DCD tag UUID, TSP TE-state reset, and IDE re-key / flush operate on grouping metadata, residue, or the link; none is dest-local grant identity.
6. A CXL P2P security token with nonce and revocation version (CN122513106A) is bound to an authorized address range, not to a dest-assigned G independent of address, and does not fail-close the source on the memory transaction.
7. InfiniBand Type-2 memory windows with SEND_WITH_INV invalidate a reusable R_Key on a send-invalidate control path; the source still posts RDMA with the key.
8. A PASID Stop Marker drains page requests of an old incarnation before PASID reuse; it is not dest-issued write tickets on fabric stores. If G is only an incarnation of the address, this distinction fails.
9. Per-operation destination address-range registers (US 10,185,680) authorize one DMA into a start/end range and then clear; they are not grant identity independent of address, and not source fail-closed fabric forward.

Novelty

The present invention does not authorize a write by a host-visible address, a translation completion, a link key, a Fabric-Manager reusable page or range key, a fencing value the destination only checks, a nonce token bound to a range, or a remaining-count on one reusable ticket. The destination assigns an allocation-lifecycle identity **G** that is not the address; authorization of a single operation is a dest-issued write ticket **DCC**; the source is fail-closed on the memory transaction; the destination drains every ticket issued under **G** before the same host-visible address may be bound to a successor grant. The novelty is this **combination**, not any one of the following steps. The system:

1. Maintains, at the destination memory boundary, a grant record whose identity **G** is assigned by the destination. **G** names an **allocation lifecycle**. **G** is not a host-visible address (HPA/DPA), not a device physical address used as the grant, not a link-security key (IDE), not a requester ID, not an FM-installed reusable page or range key, not a translation result, and not a counter or epoch on the address. **G** is not the authorization of a single operation; **DCC** is. The same host-visible address after release and re-add receives **G-prime**, not **G**.
2. For a requester context **C** and input identifier **Ain**, looks up the live grant **G** and may issue a dest-issued write ticket **DCC** bound to (**C**, **Ain**, **G**, **N**). Each **N** is its own nonce: individually non-transferable, bound to that tuple and a sequence value, and terminal after accept, abort, or expire. **N** is not a remaining-count on one reusable **DCC**.
3. At the source boundary, does not place the corresponding **memory transaction** on the fabric, and does not mark it committable, unless the source holds an unconsumed **DCC** for that operation. That gate is on the memory transaction, not on a send-invalidate control path.
4. On receipt, verifies **DCC**, checks that **G** is still the live grant, and consumes **N** exactly once to a terminal state; only then does the destination **accept the operation as an authorized write**. This covers DRAM, CXL memory, persistent memory, and device memory — not a commit posted to one media type only. An IDE-valid, ATS-translated, or TE-state-reset request is not sufficient. A ticket that names a dead **G** is not accepted as an authorized write, even if the address bits match the new backing.
5. Does not bind a successor grant **G-prime** to that same host-visible address unless **Outstanding(G) = 0**: every **N** issued under **G** is in a terminal state.
6. May cover pipeline round-trip by issuing several distinct single-use **N** under live **G**. Each **N** is a separate nonce. One reusable **DCC**, or a remaining-count on one **DCC**, is ATS and is forbidden.

Grant-record states (UNBOUND, LIVE, DRAINING) and ticket states (ISSUED, RESERVED, COMMITTED, ABORTED, EXPIRED) are a **written description** of how consume-once and the reuse barrier may be carried out. They are **not** claimed as a two-phase commit protocol and are **not** a second invention. LIVE only while **G** is the live grant. DRAINING until **Outstanding(G) = 0**, then the destination assigns **G-prime**. The present disclosure is a technology concept (TRL 2). Independents are not experimentally shown. No laboratory protocol model is claimed as a reduction to practice. As a consequence of the combination, the following refusals are structural, not policy: **DCC** is not issued; the memory transaction is not forwarded and is not marked committable; **N** is not consumed to a terminal state; the operation is not accepted as an authorized write; **G-prime** is not bound to the same host-visible address while **Outstanding(G)** is not zero.

Synergistic technical effects

The combination of elements in the present invention produces synergistic technical effects that would not be predicted by combining a cached ATS completion with a MAC over host physical address, a Fabric-Manager reusable key, or a destination-only fence. Each effect below is a property of the **combination**. No single check is the invention.

Effect 1: Destination-assigned **G independent of address, with **DCC** bound to **G**, so address reuse does not honor old tickets.**

The obvious pairing is ATS plus a MAC or permission table keyed by HPA (US 10,949,358 and related Intel ATS-security patents). That pair still keys authorization by the address. After DCD-style release and re-add at the same HPA, an old MAC/HPA can still look valid. Combining ATS with an FM-installed R-Key or L_exp still leaves a reusable key the destination only checks.

The present system binds DCC to dest-assigned **G**, which is not the address and not an epoch on the address. The same host-visible address later bound to **G-prime** does not honor a ticket that names dead **G**, even if the address bits match. That split — address may be reused; grant identity may not — is not predicted by “ATS plus MAC(HPA)” or by “opaque key, destination validates.”

Effect 2: Destination-issued consume-once tickets plus source fail-closed on the memory transaction.

The obvious destination-side reaction is a fence or fencing value (WriteGuards): storage rejects a delayed write after ownership change, while the source still issues the write. SEND_WITH_INV invalidates a reusable R_Key on a send-invalidate path; the source still posts the memory operation. A range-bound nonce token (CN122513106A) lets the destination reject a reused nonce; it does not fail-close the source on the memory transaction.

The present system issues DCC at the destination and **fails closed at the source**: absent an unconsumed DCC bound to (C, Ain, G, N), the source does not place the memory transaction on the fabric and does not mark it committable. Consume-once is dest-side (N terminal after accept, abort, or expire). Source no-forward is not predicted by a dest-only check, and dest-issued **G** is not predicted by a reusable page key the dest only validates.

Effect 3: Outstanding(G) = 0 before G-prime at the same host-visible address.

The obvious reuse hygiene is IDE re-key / flush (CVE-2025-9614), TSP TE-state reset on add, PASID Stop Marker drain of page requests, or clearing dest range registers after one DMA. Each handles a different object: a link key, residue on a range, a page-request incarnation, or a per-operation address window. None is dest-local allocation-lifecycle identity plus dest-issued tickets plus source no-forward. The present system will not bind **G-prime** at that same host-visible address unless every **N** issued under **G** is in a terminal state. Pipeline round-trip is covered by several distinct single-use **N** under live **G**, not by a remaining-count on one DCC. IDE-valid, ATS-translated, or TE-state-reset is not sufficient for an authorized write. Ticket states that implement consume-once and this drain are enablement of those limitations, not a two-phase commit claimed as the invention. Grant identity **G**, nonce **N**, and Outstanding(**G**) remain **control parameters of the gate**. The product is whether the source forwards, whether the destination accepts an authorized write, and whether the destination binds **G-prime**.

Definitions

The following terms are used throughout this specification and claims with the specific meanings defined below.

"Destination memory boundary" means processing circuitry and memory at the responder / device / pooled-memory side that maintains the grant record, issues DCC, consumes **N**, accepts or refuses an authorized write, and binds or refuses **G-prime**. It is not a host IOMMU translation agent and not a Fabric Manager.

"Source boundary" means processing circuitry and memory at the requester side that does or does not place a memory transaction on the fabric and does or does not mark it committable. The source-boundary gate is on the **memory transaction**, not on a send-invalidate control path.

"Host-visible address" means the address bits used by the requester to name backing (including HPA and, where mapped, a device physical address used as that name). It is not grant identity **G**.

"Grant identity G" means a destination-assigned identifier of an **allocation lifecycle**. **G** is not HPA, DPA, an IDE key, a requester ID, an FM-installed reusable page or range key, a translation result, or a counter or epoch on the address. **G** is not the authorization of a single operation.

"Grant record" means dest-local state associating live **G** with the current allocation. Written enablement: UNBOUND → LIVE → DRAINING → UNBOUND, then dest assigns **G-prime**. These names describe consume-once and the reuse barrier; they are not a two-phase commit claimed as the invention.

"Requester context C" means the dest-visible identity of the requester for ticket issue (for example a process, PASID, or equivalent context the destination uses to look up live **G**). **C** is not **G**.

"Input identifier Ain" means the dest-visible identifier of the operation for which a ticket may be issued. **Ain** is not **G** and is not a remaining-count.

"Write ticket DCC" means a dest-issued, single-use authorization bound to (C, Ain, G, N). Authorization

of a single operation is DCC, not G. A remaining-count on one DCC is forbidden.

"**Nonce N**" means the consume-once identifier of one DCC. Each N is its own nonce: non-transferable, sequence-bound, and terminal after accept, abort, or expire. N is not a remaining-count on one reusable DCC.

"**Unconsumed DCC**" means a write ticket whose N has not yet reached a terminal state. Absent an unconsumed DCC, the source boundary does not forward.

"**Authorized write**" means the destination has verified DCC, confirmed G is still the live grant, consumed N exactly once, and accepted the operation as a write into the allocation named by G. It is not "posted to one media type" only, not an ATS translation, and not an IDE-valid flit.

"**Outstanding(G)**" means the set of N issued under G that are not yet in a terminal state. $\text{Outstanding}(G) = 0$ if and only if every such N is accepted, aborted, or expired.

"**Successor grant G-prime**" means a new dest-assigned allocation-lifecycle identity bound to the same host-visible address after G. The destination does not bind G-prime unless $\text{Outstanding}(G) = 0$.

"**Live grant**" means the G currently bound at the destination for that allocation. A ticket that names a dead G is not accepted as an authorized write, even if the address bits match the new backing.

"**Source fail-closed**" means the default at the source boundary is not to place the memory transaction on the fabric and not to mark it committable, unless an unconsumed DCC for that operation is held.

"**Consume-once**" means the destination takes N to a terminal state exactly once. Written enablement of N: ISSUED $\rightarrow$ RESERVED $\rightarrow$ COMMITTED or ABORTED; or ISSUED $\rightarrow$ EXPIRED. Reissue of G-prime only when $\text{Outstanding}(G) = 0$.

"**Ticket window**" means several distinct single-use N issued under live G so that pipeline round-trip may be covered. Each N remains a separate nonce. One reusable DCC, or a prepaid remaining-count on one DCC, is ATS and is not a ticket window.

"**Field of use**" means fabric-attached, device-attached, pooled, and disaggregated memory, including CXL- and PCIe-connected memory. Field of use is not a limitation that makes G an address, an R-Key, or an IDE key.

5.Objective(s) of Invention

1. To provide a computer-implemented write-authorization gate in a memory system in which the destination assigns an allocation-lifecycle grant identity **G** that is independent of the host-visible address and independent of the address-translation result, such that G is not HPA, DPA, an IDE key, a requester ID, an FM-installed reusable page or range key, or a counter or epoch on the address.
2. To issue, at the destination, a consume-once write ticket **DCC** bound to (C, Ain, G, N), where each **N** is its own nonce, non-transferable, sequence-bound, and terminal after accept, abort, or expire, and is not a remaining-count on one reusable DCC.
3. To fail-close the source boundary on the **memory transaction**: absent an unconsumed DCC for that operation, the source does not place the transaction on the fabric and does not mark it committable. The gate is not a send-invalidate control path.
4. To accept an operation as an **authorized write** at the destination only after DCC is verified, G is still the live grant, and N is consumed exactly once; an IDE-valid, ATS-translated, or TE-state-reset request is not sufficient.
5. To refuse as an authorized write a ticket that names a dead G, even if the address bits match the new backing after the same host-visible address is bound to a successor grant.
6. To bind a successor grant **G-prime** to that same host-visible address only when **Outstanding(G) = 0**: every N issued under G is in a terminal state.
7. To cover pipeline round-trip by issuing several distinct single-use N under live G, each N remaining a separate nonce, and to forbid a cached multi-use DCC or a prepaid remaining-count on one DCC.
8. To treat grant identity G, nonce N, and $\text{Outstanding}(G)$ as embedded control parameters of those gates, not as the product of the invention; the product is whether the source forwards, whether the destination accepts an authorized write, and whether the destination binds G-prime.

9. To apply the gate across DRAM, CXL memory, persistent memory, and device memory as field of use, without claiming a two-phase commit protocol, an address-translation service, a link-encryption scheme, or a rule that delayed writes must be prevented.

6. Working principle of the invention (in brief)

The system operates as a seven-stage write-authorization pipeline at a **source boundary** and a **destination memory boundary**, operatively coupled over a fabric or device interconnect. Fabric-attached, device-attached, pooled, and disaggregated memory is the field of use. The invention is dest-assigned allocation-lifecycle grant identity **G** and a source-fail-closed write-authorization gate: the source does or does not place a memory transaction on the fabric; the destination does or does not accept an **authorized write**; the destination does or does not bind **G-prime** to the same host-visible address.

Stage 1: GRANT BIND. The destination maintains a grant record and assigns identity **G**. **G** names an **allocation lifecycle**. **G** is not a host-visible address (HPA/DPA), not a link-security key (IDE), not a requester ID, not an FM-installed reusable page or range key, not a translation result, and not a counter or epoch on the address. **G** is not the authorization of a single operation. Written enablement of the grant record: UNBOUND → LIVE. LIVE only while **G** is the live grant. The same host-visible address, after a later rebind, receives **G-prime**, not **G**.

Stage 2: TICKET ISSUE. For a requester context **C** and input identifier **Ain**, the destination looks up the live grant **G** and may issue a dest-issued write ticket **DCC** bound to (**C**, **Ain**, **G**, **N**). Each **N** is its own nonce: non-transferable, sequence-bound, and terminal after accept, abort, or expire. **N** is not a remaining-count on one reusable **DCC**. Authorization of a single operation is **DCC**, not **G**. If the destination does not issue **DCC**, the source remains fail-closed.

Stage 3: SOURCE GATE. Absent an unconsumed **DCC** for that operation, the source boundary does not place the corresponding **memory transaction** on the fabric and does not mark it committable. That gate is on the memory transaction, not on a send-invalidate control path.

Stage 4: DEST ACCEPT. On receipt, the destination verifies **DCC**, checks that **G** is still the live grant, and consumes **N** exactly once to a terminal state. Only then does it **accept the operation as an authorized write**. This covers DRAM, CXL memory, persistent memory, and device memory — not a commit posted to one media type only. An IDE-valid, ATS-translated, or TE-state-reset request is not sufficient. A ticket that names a dead **G** is not accepted, even if the address bits match the new backing. Written enablement of **N**: ISSUED → RESERVED → COMMITTED or ABORTED; or ISSUED → EXPIRED.

Stage 5: TICKET WINDOW. Pipeline round-trip may be covered by issuing several distinct single-use **N** under live **G**. Each **N** remains a separate nonce. One reusable **DCC**, or a prepaid remaining-count on one **DCC**, is ATS and is forbidden.

Stage 6: DRAIN. The destination does not bind a successor grant **G-prime** to that same host-visible address unless **Outstanding(G) = 0**. Written enablement: LIVE → DRAINING until every **N** issued under **G** is in a terminal state. DRAINING is destination drain of dest-issued tickets. It is not a PASID Stop Marker, not an IDE re-key, and not a TE-state reset on add.

Stage 7: REBIND. When **Outstanding(G) = 0**, the destination may assign **G-prime** at the same host-visible address. Tickets that name **G** are dead. Address bits may match the new backing; grant identity does not. The source remains fail-closed for operations that do not hold an unconsumed **DCC** bound to **G-prime**. Grant-record states and ticket states are a **written description** of how consume-once and the reuse barrier may be carried out. They are **not** claimed as a two-phase commit protocol and are **not** a second invention. **G**, **N**, and **Outstanding(G)** remain control parameters of the gates, not the product. The present disclosure is a technology concept (TRL 2). Independents are not experimentally shown.

7. Description of the invention in detail (Include drawing and or photograph as needed)

7.0 Scope of this description

This section describes the invention as a **technology concept (TRL 2)**. Independents are not experimentally shown. No laboratory protocol model is claimed as a reduction to practice. Figures 1–6 are engineering drawings of the formulated apparatus. They are not oscilloscope traces, not kernel logs, and not a CXL specification edit.

Fabric-attached, device-attached, pooled, and disaggregated memory, including CXL- and PCIe-connected memory, is the **field of use**. The invention is dest-assigned allocation-lifecycle grant identity **G** and a source-fail-closed write-authorization gate. Grant-record states and ticket states named below are a **written description** of how consume-once and the reuse barrier may be carried out. They are **not** claimed as a two-phase commit protocol.

G , N , the tuple (C, A_{in}, G, N) , and Outstanding(G) are **embedded control parameters of the gate**. The product is whether the source places a memory transaction on the fabric (or marks it committable), whether the destination accepts an **authorized write**, and whether the destination binds **G-prime** to the same host-visible address.

7.1 Figure 1 — Apparatus: source boundary and destination memory boundary

Figure 1 shows the apparatus. A requester context **C** names an operation by an input identifier **A_{in}** and by host-visible address bits. A **source boundary** and a **destination memory boundary** are operatively coupled over a fabric or device interconnect. Each boundary comprises processing circuitry coupled to a memory.

The destination maintains a grant record and **assigns G**. G names an allocation lifecycle. G is not HPA, not DPA used as the grant, not an IDE key, not a requester ID, not an FM-installed reusable page or range key, not a translation result, and not a counter or epoch on the address. Authorization of a single operation is not G ; it is a dest-issued write ticket **DCC** bound to (C, A_{in}, G, N) .

The source boundary does not place the corresponding **memory transaction** on the fabric, and does not mark it committable, unless it holds an unconsumed DCC for that operation. That gate is on the memory transaction. It is not a send-invalidate control path (SEND_WITH_INV).

On receipt, the destination verifies DCC, checks that G is still the live grant, and consumes N exactly once. Only then does it accept the operation as an authorized write. This covers DRAM, CXL memory, persistent memory, and device memory — not a commit posted to one media type only. An IDE-valid, ATS-translated, or TE-state-reset request is not sufficient.

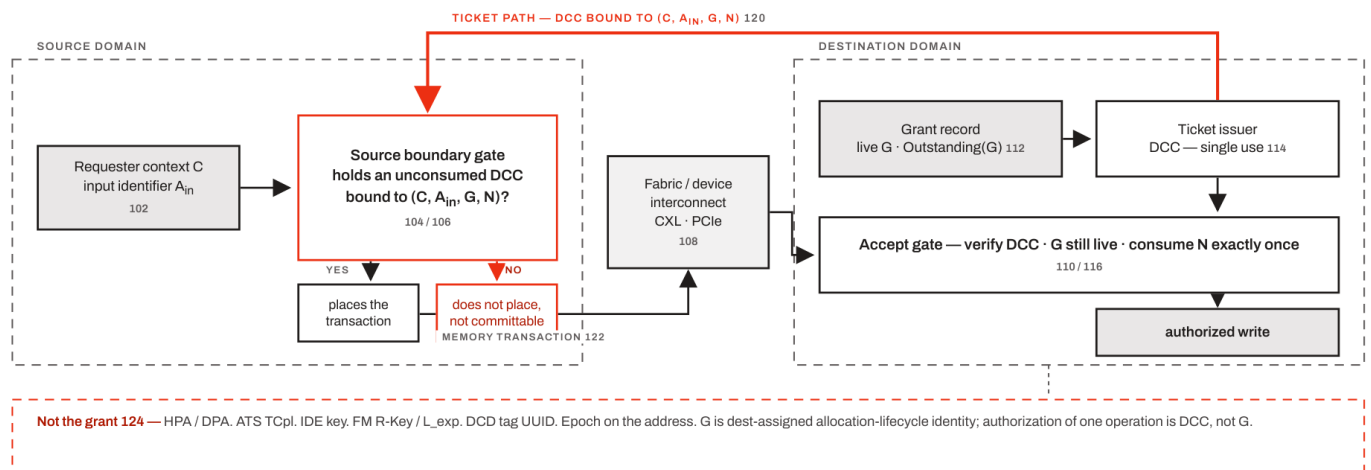


Figure 1. Source boundary and destination memory boundary. G is not the address.

7.2 Figure 2 — Technical effect: three structural control points

Figure 2 states the product. The combination yields three structural control points. They are independents of the disclosure, not a policy that delayed writes must be prevented.

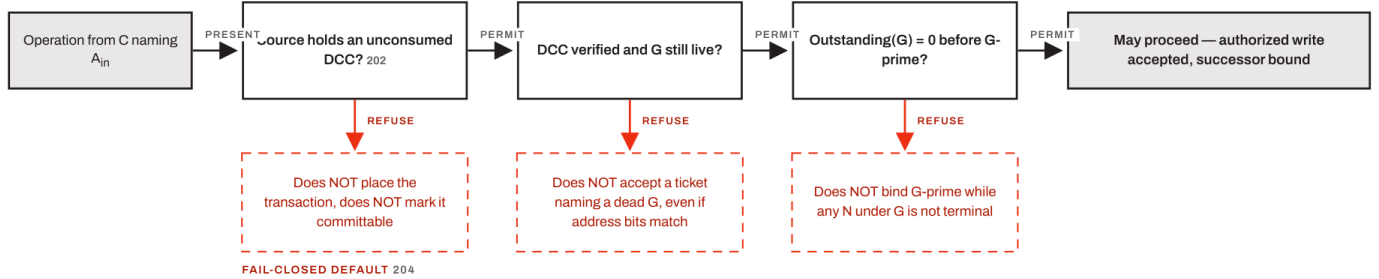


Figure 2. Independents are structural refusals of forward, accept, and rebind.

1. DCC is not issued at the destination.
2. The memory transaction is not placed on the fabric and is not marked committable.
3. N is not consumed to a terminal state.
4. The operation is not accepted as an authorized write.
5. G-prime is not bound to the same host-visible address while Outstanding(G) is not zero.

If the source holds an unconsumed DCC, it **may** place the memory transaction. If the destination verifies DCC, G is live, and N is consumed once, it **may** accept an authorized write. If Outstanding(G) = 0, it **may** bind G-prime at that same host-visible address. The refusals in Figure 2 (red) are the fail-closed defaults.

7.3 Figure 3 — Host-visible address is not grant identity

Figure 3 is the distinction the invention owns: **the address is not the grant**. Pooled and dynamic-capacity memory may release backing and later add capacity that the host maps at the same host-visible address. Linux CXL DCD (2026 RFC) groups extents that share a non-null **tag UUID** and orders them by **seq_num**; release is keyed by (DPA range, tag), not by a dest-assigned write-ticket. Tag UUID is mailbox grouping metadata. It is not dest-local G and not a consume-once DCC.

After dest binds **G-prime** at HPA A, a ticket DCC(C, A_{in}, G, N) names **dead G**. The destination does not accept it as an authorized write even if the address bits still equal A. US 10,949,358 authenticates a device-supplied HPA with a MAC and an invalidation-tracking table. After reuse of the same HPA, an old MAC/HPA can still look valid. Binding DCC to dest-assigned G, which is not the address and not an epoch on the address, is what makes the old ticket fail.

If G were only an incarnation or epoch of the address, this figure would collapse to a PASID Stop Marker or a DCD generation counter. G is dest-assigned allocation-lifecycle identity **independent of** the address.

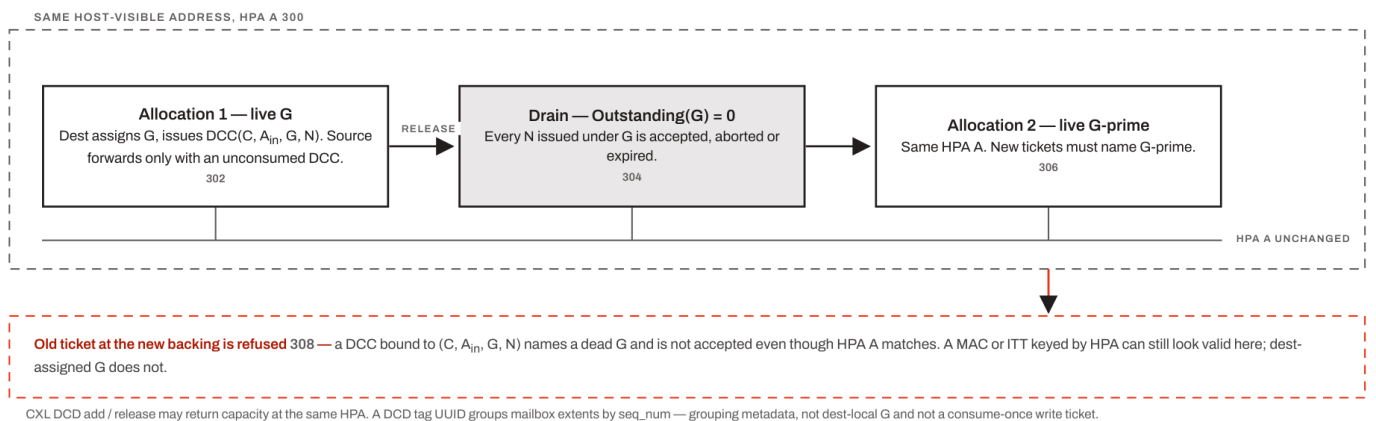


Figure 3. Address bits may be reused. Grant identity G is not reused with them.

7.4 Figure 4 — Consume-once N versus cached ATS completion

Figure 4 distinguishes consume-once N from Address Translation Services. A device sends a Translation Request (AT=01). The translation agent returns a Translation Completion (TCpl / CplD) carrying HPA and permissions. The device stores that completion in an Address Translation Cache. Later memory requests

use $AT=10$ with the cached HPA and **need no new completion** until invalidation. One TCpl is therefore multi-use.

A remaining-count on one DCC, or one reusable DCC used for many operations, is the same shape as a cached TCpl. It is ATS and is forbidden. In this invention each N is its own nonce: non-transferable, sequence-bound, and terminal after accept, abort, or expire. Pipeline round-trip may be covered by issuing **several distinct** single-use N under live G. That is a ticket window, not a prepaid counter on one ticket. Translation is not an authorized write. An ATS-translated request that does not carry an unconsumed DCC naming the live G is not accepted as an authorized write at the destination, and is not placed on the fabric by a fail-closed source.

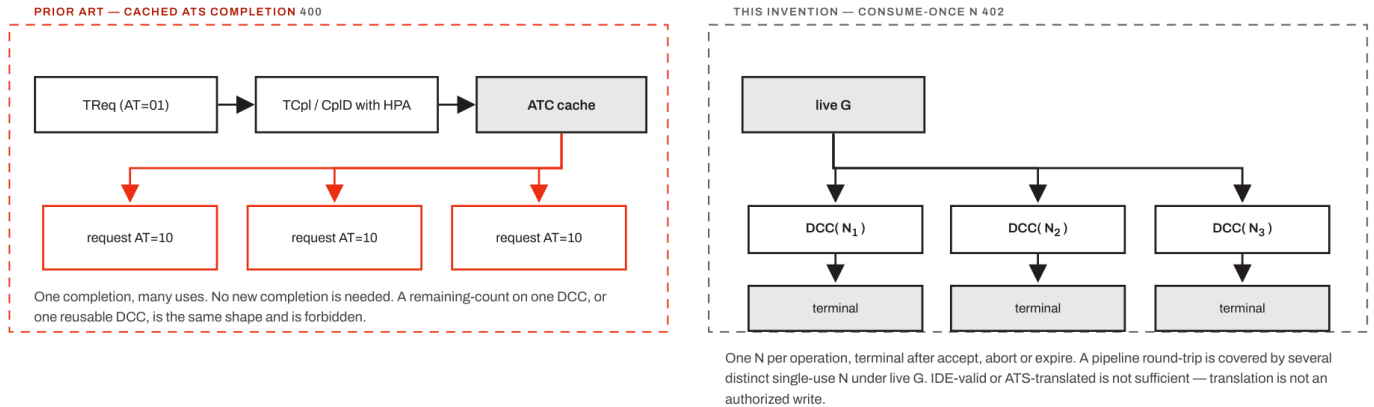


Figure 4. Cached TCpl authorizes later requests. Each claimed ticket N does not.

7.5 Figure 5 — Source fail-closed versus destination-only check

Figure 5 is the source-side half of the combination. WriteGuards (OSDI 2026) is a storage fencing value tied to key-range ownership: the store rejects a delayed write; the source **still issues** the write. InfiniBand Type-2 memory windows with **SEND_WITH_INV** invalidate a reusable R_Key on a send-invalidate path; the source still posts RDMA with the key. CN122513106A lets a CXL P2P destination reject a reused nonce or an old revocation version VN on a token bound to an **address range**; it does not fail-close the source on the memory transaction. Gen-Z R-Key and Space-Control L_exp are FM-installed reusable keys the destination only checks.

This invention issues DCC at the destination **and** fails closed at the source. Absent an unconsumed DCC bound to (C, Ain, G, N), the source does not place the memory transaction on the fabric and does not mark it committable. Consume-once is dest-side. No-forward is source-side. A dest-only fence is not this combination. An opaque key the dest only validates is not this combination.

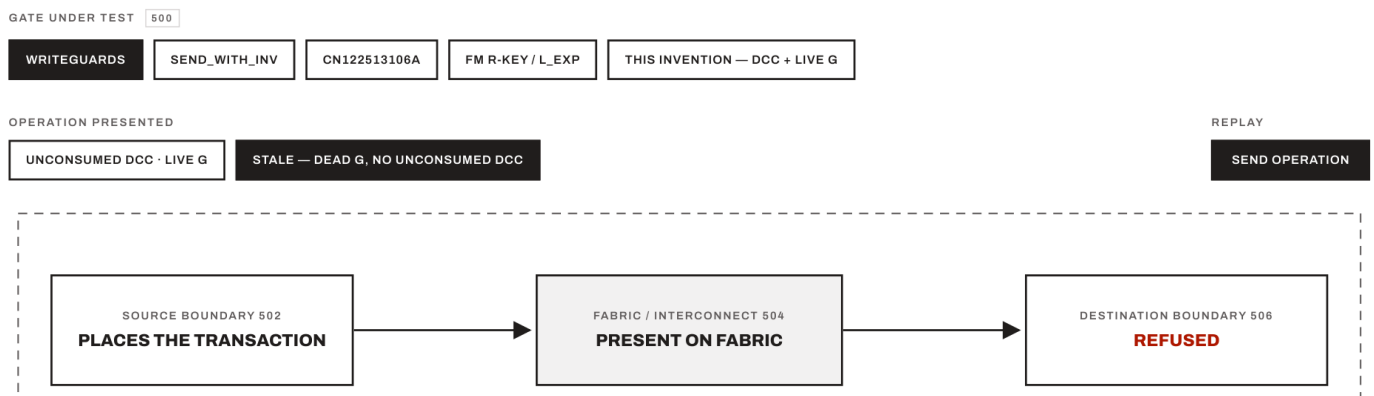


Figure 5. A destination-only fence is not a source-fail-closed memory-transaction gate.

7.6 Figure 6 — Drain Outstanding(G) before G-prime

Figure 6 is the reuse barrier. The destination does not bind **G-prime** to that same host-visible address unless **Outstanding(G) = 0**: every N issued under G is in a terminal state. Written enablement of the grant record: **UNBOUND** → **LIVE** → **DRAINING** → **UNBOUND**, then dest assigns G-prime. LIVE only while

G is the live grant. DRAINING until Outstanding(G) = 0.

Written enablement of N: ISSUED → RESERVED → COMMITTED or ABORTED; or ISSUED → EXPIRED. These names describe consume-once and the drain. If Form B sold ISSUED / RESERVED / COMMITTED as the invention, the disclosure would be two-phase-commit folklore. They are enablement of limitations already stated, not a second invention.

This drain is not IDE re-key / flush on TDI rebind (CVE-2025-9614 Delayed Posted Redirection). It is not CXL TSP TE-state reset on add, which clears residue and does not say the range is the same grant. It is not a PASID Stop Marker, which drains page requests of an old PASID incarnation. It is not DCD tag UUID / seq_num. It is not clearing destination DMA range registers after one operation (US 10,185,680).

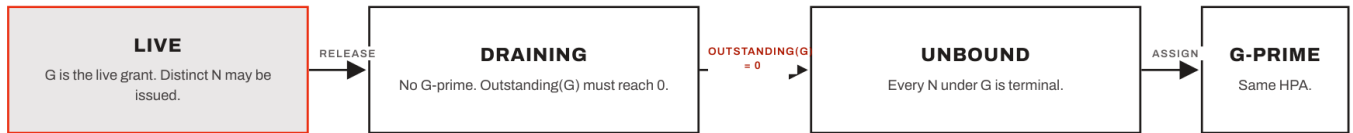


Figure 6. Dest will not bind G-prime at that address until every dest-issued N under G is terminal.

7.7 Grant record (written enablement)

The grant record is dest-local state associating live G with the current allocation. The destination **assigns** G; a Fabric Manager does not install G as a reusable page key. C is dest-visible requester context (for example a process or PASID the destination uses to look up live G). C is not G. Ain identifies the operation for ticket issue. Ain is not G and is not a remaining-count.

While LIVE, the destination may issue several distinct N under G (ticket window). When the allocation is to be released, the destination enters DRAINING: it does not bind G-prime, and it does not honor new tickets that would outlive the drain rule. When Outstanding(G) = 0, the destination may unbind G and assign G-prime at the same host-visible address. Tickets that name G are then dead.

7.8 Write ticket DCC and ticket window (written enablement)

DCC is dest-issued, single-use, and bound to (C, Ain, G, N). Each N is a nonce of that ticket. Transfer of to another context, reuse of N after a terminal state, or a remaining-count on one DCC is outside this disclosure. Sequence-binding of N is so that tickets under live G are distinguishable; it is not a remaining-count.

On dest accept: DCC verifies; G is still live; N is consumed exactly once; the operation is accepted as an authorized write into the allocation named by G. On dest refuse: any of those checks fails, including a ticket that names dead G after G-prime is bound at the same address bits.

7.9 Illustrative sequence at one host-visible address

Item	First allocation	During drain	Successor allocation
Host-visible address	HPA A	HPA A	HPA A
Grant identity	G (LIVE)	G (DRAINING)	G-prime (LIVE)
Write ticket	DCC (C, Ain, G, N)	Every N under G driven to a terminal state	DCC (C, Ain, G-prime, N')
Old DCC (C, Ain, G, N)	Unconsumed: source may forward	Must become terminal before G-prime	Refused: names dead G, even if HPA A matches
ATS TCpl / IDE / tag UUID	Not sufficient for authorized write	Not the drain of Outstanding(G)	Not sufficient; not G-prime

The table is an illustrative sequence, not a laboratory trace and not a claimed numeric result. Values such as “HPA A” are labels. The same sequence applies to device physical addresses used as the host-visible name, and to CXL DCD add after release at that name.

7.10 Boundaries of this description

This description does **not** claim a two-phase commit protocol, an address-translation service, a link-encryption scheme, a Fabric-Manager reusable R-Key or L_exp, a destination-only fencing check, a nonce token bound to an address range, a remaining-count on one reusable ticket, or a rule that delayed writes must be prevented.

If G is described only as an epoch or counter on the address, the disclosure collapses to fencing / PASID incarnation / DCD generation. If DCC is described as a cached multi-use completion, it collapses to ATS. If the source still forwards and the destination only checks a fence or a range-bound nonce, dest-issued G plus source fail-closed is not described. Those boundaries are part of the written description so that the combination in Figures 1–6 is what is disclosed.

8.Experimental validation results:

8.1 Status of validation

This disclosure is a **technology concept (TRL 2)**. **No experimental validation results are presented.** Independents are not experimentally shown. The written description in heading 7, including Figures 1–6, is not evidence of function and is not a laboratory protocol model.

8.2 What is not presented

This section does **not** present protocol traces, kernel logs, QEMU runs, CXL specification edits, false-positive or false-negative percentages, ablation tables, or stress-test counts. No such campaign is part of this TRL 2 disclosure. Inventing laboratory metrics would misstate the readiness level.

8.3 Drawings versus traces

Figures 1–6 are engineering drawings of the formulated apparatus. They are not oscilloscope captures, not fabric traces, and not a reduction to practice. Grant-record states (UNBOUND, LIVE, DRAINING) and ticket states (ISSUED, RESERVED, COMMITTED, ABORTED, EXPIRED) named in heading 7 are a written description of how consume-once and the reuse barrier **may** be carried out. They are not experimental records.

8.4 Readiness outside this disclosure

Experimental proof of concept (**TRL 3**) and laboratory validation (**TRL 4**) lie **outside** this disclosure. This heading therefore contains no pass/fail table and no claimed accuracy figure.

9.What aspect(s) of the invention need(s) protection?

The aspects that need protection are the **combination** below as a destination-assigned allocation-lifecycle grant identity and a source-fail-closed write-authorization gate in a memory system: the source places a memory transaction on the fabric, or it does not; the destination accepts an authorized write, or it does not; the destination binds a successor grant at the same host-visible address, or it does not. The novelty is not a cached translation completion, not a MAC over host physical address, not an opaque key the destination only checks, not a two-phase commit protocol, and not a rule that delayed writes must be prevented. Fabric-attached memory is the field of use.

Outside the scope of protection sought: a MAC or invalidation table keyed by host physical address; a Fabric-Manager reusable R-Key or L_exp; a destination-only fencing check with the source still forwarding; a cached or reusable DCC; a remaining-count on one DCC; grant identity G described only as an epoch or counter on the address; a nonce token bound to an authorized address range; IDE re-key or flush on rebind; ISSUED / RESERVED / COMMITTED claimed as the invention; a load-only translation paired with a store-posting authorization; and a result-only rule that stale or delayed DMA must be prevented. Numerical pipeline depths and fabric names are field of use, not limitations that turn G into an address.

Independent 1 (System)

A computer-implemented write-authorization system for a memory system, comprising a source boundary and a destination memory boundary, each comprising processing circuitry coupled to a memory, the source boundary and the destination memory boundary being operatively coupled over a fabric or device interconnect, the circuitry being configured to:

- (a) maintain, at the destination memory boundary, a grant record whose identity G is assigned by the destination, G naming an allocation lifecycle, G not being a host-visible address, not a device physical address used as the grant, not a link-security key, not a requester identifier, not a Fabric-Manager-installed reusable page or range key, not a translation result, and not a counter or epoch on the address, G not being the authorization of a single operation, the same host-visible address after release and re-add receiving a successor grant G -prime and not G ;
- (b) for a requester context C and an input identifier A_{in} , look up the live grant G at the destination and may issue a dest-issued write ticket DCC bound to (C, A_{in}, G, N) , each N being its own nonce, individually non-transferable, bound to that tuple and a sequence value, and terminal after accept, abort, or expire, N not being a remaining-count on one reusable DCC, authorization of a single operation being DCC and not G ;
- (c) at the source boundary, refrain from placing the corresponding memory transaction on the fabric and refrain from marking it committable unless the source boundary holds an unconsumed DCC for that operation, that gate being on the memory transaction and not on a send-invalidate control path;
- (d) at the destination memory boundary, verify DCC, check that G is still the live grant, consume N exactly once to a terminal state, and only then accept the operation as an authorized write, an IDE-valid request, an ATS-translated request, or a TE-state-reset of a range not being sufficient for that accept, a ticket that names a dead G not being accepted as an authorized write even if the address bits match the new backing, the accept covering DRAM, CXL memory, persistent memory, and device memory and not being limited to a commit posted to one media type;
- (e) refrain from binding successor grant G -prime to that same host-visible address unless $\text{Outstanding}(G) = 0$, $\text{Outstanding}(G) = 0$ meaning that every N issued under G is in a terminal state, namely accepted as an authorized write, aborted, or expired; and
- (f) optionally cover pipeline round-trip by issuing several distinct single-use N under live G , each N remaining a separate nonce, one reusable DCC or a remaining-count on one DCC being forbidden.

Independent 2 (Method)

A method of operating a write-authorization gate in a memory system, the method comprising:

- (a) assigning, by a destination memory boundary, a grant identity G of an allocation lifecycle, G not being a host-visible address, not a device physical address used as the grant, not a translation result, not a link-security key, not a requester identifier, not a Fabric-Manager-installed reusable page or range key, and not a counter or epoch on the address, G not being the authorization of a single operation, the same host-visible address after release and re-add receiving G -prime and not G ;
- (b) issuing, by the destination, a dest-issued write ticket DCC bound to (C, A_{in}, G, N) for a requester context C and input identifier A_{in} , each N being its own nonce and not a remaining-count on one reusable DCC;
- (c) fail-closing a source boundary on the memory transaction, whereby the source does not place that transaction on the fabric and does not mark it committable unless it holds an unconsumed DCC for that operation, the gate not being a send-invalidate control path;
- (d) verifying DCC at the destination, checking that G is still the live grant, consuming N exactly once, and only then accepting the operation as an authorized write, an IDE-valid, ATS-translated, or TE-state-reset request not being sufficient, a ticket that names a dead G not being accepted even if the address bits match the new backing;
- (e) not binding a successor grant G -prime to that same host-visible address unless $\text{Outstanding}(G) = 0$; and
- (f) issuing, when a ticket window is used, several distinct single-use N under live G , each N remaining a separate nonce, a remaining-count on one DCC being forbidden.

The product of both independents is whether the source places the memory transaction (or marks it committable), whether the destination accepts an authorized write, and whether the destination binds G -prime. G , N , and $\text{Outstanding}(G)$ remain control parameters of those gates. The combination is what is

protected. No single letter (a)–(f) is the invention. A cached ATS completion plus a remaining-count does not produce Independent 1(b) and 1(c). A destination-only fence does not produce Independent 1(b) and 1(c). A nonce token bound to an address range does not produce Independent 1(a) and 1(c).

Recitation	Prior-art reply	Technical effect (gate)
1(a), 2(a)	Not HPA, not ATS TCpl, not DCD tag UUID, not an epoch on the address	The same host-visible address after re-add receives G-prime, not G
1(b), 1(d), 2(b), 2(d)	Not a remaining-count, not a range-bound nonce token (CN122513106A)	Dest accepts an authorized write only after consume-once N while G is live
1(c), 2(c)	Not a dest-only fence (WriteGuards); not SEND_WITH_INV; not FM R-Key / L_exp	Source does not place the memory transaction without an unconsumed DCC
1(e), 2(e)	Not IDE re-key (CVE-2025-9614), not TE-state reset, not PASID Stop Marker	Dest does not bind G-prime until Outstanding(G) = 0
1(f), 2(f)	Not a cached multi-use TCpl	Pipeline round-trip uses distinct single-use N, not a prepaid counter on one DCC

Dependent 3

The system of Independent 1, wherein the several distinct N of (f) are issued only while G is the live grant, and wherein no remaining-count is maintained on any one DCC.

Dependent 4

The system of Independent 1, wherein, after G-prime is bound to the same host-visible address, the destination memory boundary does not accept as an authorized write a DCC that names G, even if the address bits match the new backing.

Dependent 5

The system of Independent 1, wherein G is independent of a dynamic-capacity tag UUID, independent of an Address Translation Services translation completion, and independent of a link integrity-and-data-encryption key.

Dependent 6

The method of Independent 2, wherein Outstanding(G) = 0 if and only if every N issued under G has been accepted as an authorized write, aborted, or expired.

Dependent 7

The system of Independent 1, wherein dest-local grant-record states UNBOUND, LIVE, and DRAINING are used as a written manner of carrying out (a) and (e), LIVE only while G is the live grant, DRAINING until Outstanding(G) = 0, those states not being protected as a two-phase commit protocol.

Dependent 8

The system of Independent 1, wherein dest-local ticket states ISSUED, RESERVED, COMMITTED, ABORTED, and EXPIRED are used as a written manner of carrying out (d), those states not being

protected as a two-phase commit protocol.

Dependent 9

The method of Independent 2, wherein the source-boundary gate of (c) is not a SEND_WITH_INV path and is not any other send-invalidate control path.

Dependent 10

The system of Independent 1, wherein the fabric or device interconnect includes a CXL or PCIe interconnect as field of use, field of use not making G an address, a Region Key, or an IDE key.

Dependent 11

The method of Independent 2, wherein an IDE-valid request, an ATS-translated request, or a TE-state-reset of a range is not treated as sufficient for an authorized write.

Dependent 12

The system of Independent 1, wherein a security token bound to an authorized address range, including a token carrying a nonce and a revocation version on that range, is not G and is not DCC bound to (C, Ain, G, N).

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	Amd. No/Date	0/00.00.0000

10. What is Technology readiness level of your invention? (Tick the appropriate TRL)

Research			Development			Deployment		
TRL 1	TRL 2	TRL 3	TRL 4	TRL 5	TRL 6	TRL 7	TRL 8	TRL 9
Basic Principles observed	Technology concept formulated	Experimental proof of concept	Technology validated in a lab	Technology validated in a relevant environment (industrially relevant in case of key enabling technologies)	Technology demonstrated in a relevant environment (industrially relevant in case of key enabling technologies)	System prototype demonstration in an operational environment	System complete and qualified	Actual system proven in an operational environment (competitive manufacturing in case of key enabling technologies, or in space)
	✓							

Justification: Technology concept formulated (TRL 2). Independents are not experimentally shown. No laboratory protocol model.

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